

Emily Cook

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SUMMARY:

Geophysicist with experience across near-surface geophysics, energy feasibility studies, seismic hazard assessment, and subsurface modelling. Proven track record managing geodatabases and field campaigns.

EDUCATION:

Binghamton University, SUNY | May 2026 | B.S. in Geophysics (3.93 GPA) | Summa Cum Laude | Dean's List

SKILLS:

Geophysical Methods: ERT, tTEM, SRT, GPR, Magnetometry, UAV/UAS, LiDAR, Hyperspectral

Software: Rayfract, ResIPy, AGI Earth Imager, ArcGIS, QGIS, Leapfrog, Thermoviewer, Pix4D, Python, Unix

Field: Orienteering, Bear Safety, First Aid, UTV Operation, Backcountry Camping

EXPERIENCE:

Geophysics & Seismology Intern

May 2026 – Present

Bureau of Reclamation (USBOR)

Denver, CO

- Field data acquisition (GPR, tTEM, ERT, SRT, MASW), processing, and 2D/3D modelling to evaluate dam safety and structural integrity of critical infrastructure.
- Processing 300+ faults, geochronology, and geophysical surveys while maintaining data integrity in federal seismic hazard geodatabase, to streamline engineering risk analysis.

Energy Researcher

May 2025 – Present

Alaska Center of Energy and Power (ACEP)

Fairbanks, AK / Remote

- Quantify greenhouse gas emissions of the Energy and Transportation sector for a comprehensive statewide policy report in collaboration with the Nature Conservancy (TNC).
- Create feasible climate solution scenarios based on emissions that benefit Alaskan constituents.
- Evaluated five Alaskan ports using Multi-Criteria Decision Analysis (MCDA) to determine the geological and economic feasibility of geothermal-driven e-fuel production for decarbonization.
- Collaborate with local stakeholders and indigenous communities to ensure energy projects prioritized energy sovereignty, culminating in a published paper and international presentations.

Geothermal Energy Intern

May 2024 – May 2025

Summer Scholars & Artists Program

Vestal, NY

- Synthesized depth-to-layer, borehole geophysical logs, and thermal datasets from 1,000+ deep well logs into centralized ArcGIS geodatabase for regional subsurface characterization.
- Constructed 3D geospatial subsurface models and cross-sections to isolate high-probability geothermal potential zones, focusing on favorable temperature, permeability, and porosity.

Undergraduate Research Assistant

January – September 2024

SUNY RF funded Basin Research Team

Vestal, NY

- Data acquisition for UAS and water-borne temperature and hydrogeophysical surveys for submarine groundwater discharge (SGD).
- Assisted geophysical field campaigns using SRT and ERT to characterize basin geometry.

Geospatial Sensing Student Researcher

August 2022 – May 2023

First-Year Immersion Program (FRI)

Vestal, NY

- Investigated SGD's role in sustainable water resource management for island communities using UAV-TIR to assess seasonality of seep, summarizing results in research paper and poster session.
- Conducted remote sensing fieldwork and cartography exercises using various geophysical sensors.

PUBLICATIONS:

Cook, E.; de Witt, M. Assessing the Feasibility of Geothermal-to-X for Sustainable Maritime Refueling in Alaska. *Clean Technol.* **2025**, 7, 115. <https://doi.org/10.3390/cleantechnol7040115>

PRESENTATIONS:

Geothermal-to-X Potential in Alaska's Maritime Industry	<i>Arctic Frontiers</i>	2026 Poster
Geothermal-to-X Potential in Alaska's Maritime Industry	<i>ACEP</i>	2025 Talk
Feasibility of Geothermal-to-X in Alaskan Arctic Shipping	<i>GEODE</i>	2025 Talk
Spatial mapping of geothermal energy variables near Binghamton, NY	<i>SSAP</i>	2024 Talk
Seasonal Variation in Detecting Submarine Groundwater Discharge Using Drone-Mounted Thermal Imaging	<i>FRI Poster Session</i>	2023 Poster
Viability of Coastal Seeps to Combat Droughts on Rapa Nui	<i>FRI Poster Session</i>	2022 Poster

HONORS & AWARDS:

Ralph Digman Award - Outstanding Earth Science Senior	2026
Dean's List (Binghamton)	All
Art of Science Competition "Best in Show" (Binghamton)	2026
Phi Beta Kappa	2025
Best Poster Award (Peer J at Arctic Frontiers)	2026
Harpur Edge High Impact Award (Binghamton)	2026, 2023
Summer Scholars & Artists Program (Binghamton)	2024
SUNY Impact Scholarship (NYS Wireless Association)	2024
Outside the Classroom Award (EF)	2023

PROJECTS:

1. **Cartographer:** Design and deploy custom mapping projects (QGIS, Leapfrog) and portfolio web pages, including custom LiDAR-art and a university submission-based chair ranking website (**2026**).
2. **Geophysics Summer School (Texas A&M):** Seismic refraction field methods; 2018 Kilauea waveform analysis and seismicity using MatLab and QGIS (**2025**).
3. **GEOL 451 and 453:** Two semester-long projects in applied electrical geophysical methods. Survey design, implementation, analysis, and processing of wood ash on soil (ERT, EMI), locating unmarked graves (GPR), locating abandoned oil and gas wells (smartphone magnetometers) (**2024-2025**).
4. **Rover Team Geology Lead:** Integration of remote sensing and hydrologic interpretation for autonomous planetary analog missions (**2023-2024**).
5. **Ecology Field Courses:** Immersive geological and ecological field study in the Everglades and Utah and Wyoming National Parks over five weeks (**2023**).
6. **Women in Science and Engineering:** Mentored by Stony Brook PhD students in STEM (**2018-2020**).

LEADERSHIP:

1. **Team Captain (Binghamton Ultimate Frisbee):** Led and managed logistics for ~35 athlete roster for travel and competitive play across East Coast; two National Championship appearances (2024-2026).
2. **Tutor:** Designed and delivered targeted STEM instruction to K-12 students, adapting complex topics for diverse learning levels and behavioral challenges (2019-2023).
3. **Race Organizer:** Team volunteer effort to organize community races of 500+ participants, fundraising for local causes raising \$30,000+.